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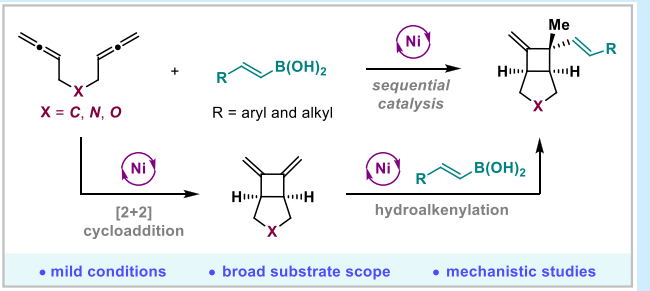
Nickel-Catalyzed Cascade Carbocyclization/Alkenylation of Bisallenes and Organoboronic Acids

Xiao-Die An, Tao Wang, Ji-Xun Guan, Yun-Xuan Tan,* and Ping Tian*

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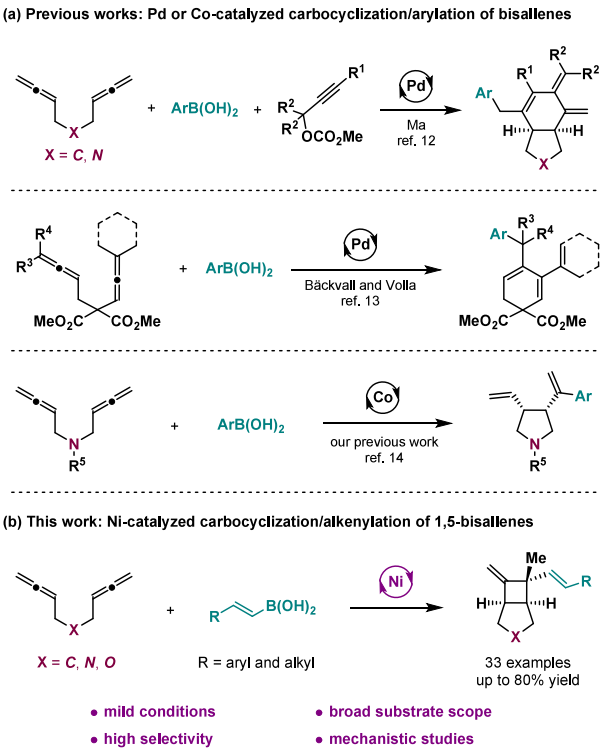
ABSTRACT: The transition metal-catalyzed cascade carbocyclization of bisallenes represents an efficient strategy for constructing versatile carbo- and heterocycles. Herein, we report a nickel-catalyzed cascade carbocyclization/alkenylation of bisallenes and organoboronic acids, enabling the synthesis of diverse bicyclo[3.2.0]heptanes with a broad substrate scope and high selectivity. Mechanistic experiments suggest that this transformation proceeds through a unique sequential nickel-catalyzed process: first, a nickel-catalyzed regioselective [2 + 2] cycloaddition of bisallenes, followed by the nickel-catalyzed hydroalkenylation of the diene intermediate with alkenylboronic acids.



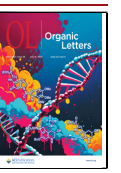
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Carbo- and heterocyclic skeletons are important structural components of bioactive natural products, drug molecules, and functional materials. The development of highly efficient, practical, and selective cyclization methods has garnered significant attention in modern organic synthesis.[1,2] Among various synthetic strategies, the transition metal-catalyzed cascade cyclization of readily accessible linear 1,*n*-bifunctional substrates has emerged as a robust approach for synthesizing complex and diverse cyclic compounds with remarkable step and atom economy.[3] Nevertheless, the transition metal-catalyzed cascade cyclization of 1,*n*-bisallenes comprising two discontinuous allene units remains relatively underexplored.[4] This is likely attributed to the difficulty in controlling the reactivity and selectivity.[5] Recent progress has been made in the transition metal-catalyzed cycloisomerization of bisallenes[6] and their cascade carbocyclization reactions with unsaturated compounds (e.g., allenes, carbon monoxide, alkenes, and alkynes).[7] However, the functionalized carbocyclization of bisallenes to produce versatile carbo- and heterocycles is highly demanding.[8−14] For instance, palladium-catalyzed bisfunctional cyclization (e.g., silastannylation, distannylation, and germastannylation),[8] borylation carbocyclization,[9] and multicomponent cascade carbocyclization of bisallenes[10] have been reported.[11] Furthermore, the transition metal-catalyzed carbocyclization/arylation of bisallenes has also been gradually implemented (Scheme 1a).[12−14] In 2009, Ma and co-workers reported a palladium-catalyzed regiose-

Scheme 1. Transition Metal-Catalyzed Carbocyclization of Bisallenes and Organoboronic Acids



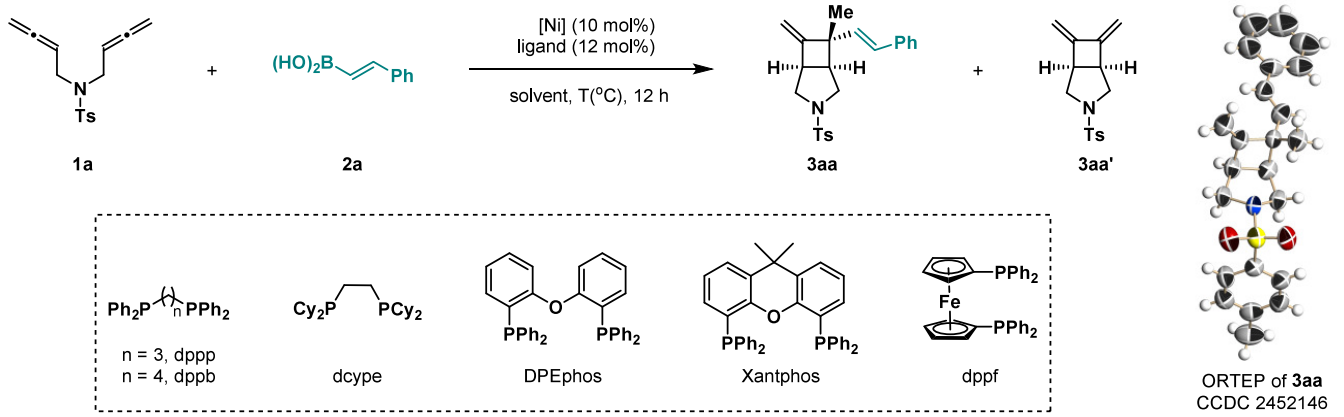
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Table 1. Reaction Development[a]



entry	[Ni]	ligand	solvent	T (°C)	yield of **3aa** (%)[b]	yield of **3aa′** (%)[b]
1	$Ni(cod)_2$	dppp	MeOH	60	0	0
2	$Ni(cod)_2$	dppb	MeOH	60	52	0
3	$Ni(cod)_2$	dcype	MeOH	60	0	0
4	$Ni(cod)_2$	DPEphos	MeOH	60	36	0
5	$Ni(cod)_2$	Xantphos	MeOH	60	53	0
6	$Ni(cod)_2$	dppf	MeOH	60	68	0
7	$Ni(acac)_2{\cdot}H_2O$	dppf	MeOH	60	23	39
8	$Ni(OAc)_2{\cdot}4H_2O$	dppf	MeOH	60	60	0
9	$Ni(cod)_2$	dppf	MeOH	50	29	0
10	$Ni(cod)_2$	dppf	MeOH	70	66	0
11	$Ni(cod)_2$	dppf	EtOH	60	54	0
12	$Ni(cod)_2$	dppf	*i*-PrOH	60	24	0
13	$Ni(cod)_2$	dppf	toluene	60	4	45
14	$Ni(cod)_2$	dppf	DCE	60	0	36
15	$Ni(cod)_2$	dppf	THF	60	6	38
16	$Ni(cod)_2$	dppf	toluene/MeOH (1/1)	60	75	0

[a]Reactions were performed with **1a** (0.1 mmol, 1.0 equiv), **2a** (0.2 mmol, 2.0 equiv), [Ni] (10 mol %), a ligand (12 mol %), and a solvent (1.0 mL) under a N_2 atmosphere for 12 h. The conversion of **1a** exceeded 95% in all cases. [b]Yield detected by ^{1}H nuclear magnetic resonance (^{1}H NMR) analysis of unpurified mixtures with CH_2Br_2 as the internal standard.

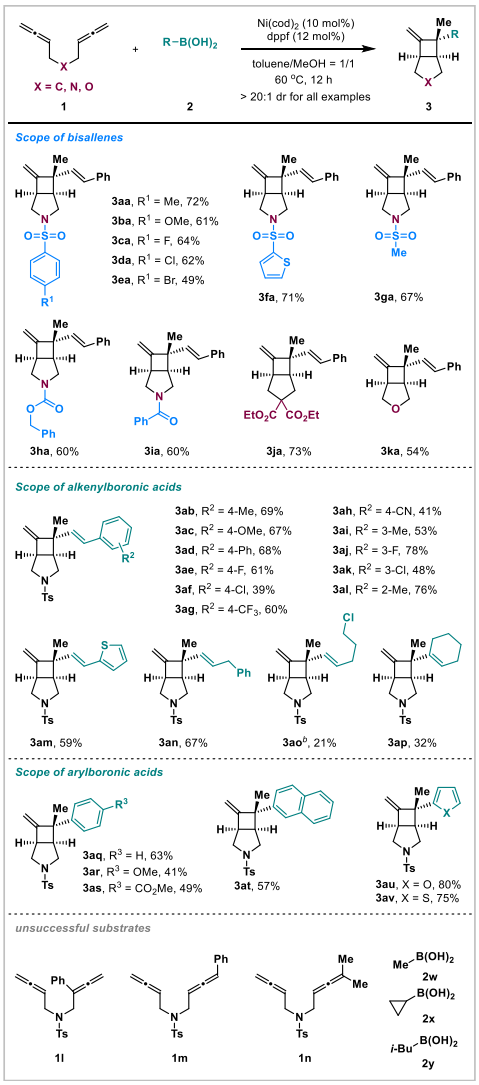
lective cascade carbocyclization of 1,5-bisallenes with propargylic carbonates and arylboronic acids, offering an efficient route to bicyclo[4.3.0]nonenes.[12] Subsequently, Bäckvall and Volla realized a palladium-catalyzed carbocyclization/arylation of 1,4-bisallenes for preparing six-membered carbocycles.[13] More recently, our group achieved the first earth-abundant cobalt- catalyzed regio- and diastereoselective carbocyclization/arylation of 1,5-bisallenes, yielding *cis*-disubstituted pyrrole derivatives.[14] Despite the aforementioned advances, the investigation of carbocyclization/alkenylation of bisallenes with alkenylboronic acids is limited, and new methods for constructing different cyclic skeletons are also in high demand.[12−14] Herein, we present a nickel-catalyzed cascade carbocyclization/alkenylation of bisallenes and alkenylboronic acids (Scheme 1b). This catalytic system accommodates not only carbon-, nitrogen-, and oxygen-linked bisallenes but also alkenyl and arylboronic acids, producing bicyclo[3.2.0]-heptanes under mild conditions with high selectivity.

The reaction conditions were optimized with bisallene **1a** and (*E*)-styrylboronic acid **2a** serving as model substrates, and the selected results are summarized in Table 1. Different commercially available bisphosphine ligands were tested in the presence of 10 mol % $Ni(cod)_2$ (Table 1, entries 1−6). To our delight, carbocyclization/alkenylation product **3aa** was successfully obtained, and no potential [2 + 2] cycloaddition product **3aa′** was detected.[6] Notably, the bisphosphine ligand with a small natural bite angle failed to generate **3aa** (Table 1, entries 1 and 3).[15] The relative configuration of **3aa** was determined by X-ray crystallography (CCDC 2452146). Then, nickel(II) catalysts were examined, but the yield could not be obviously enhanced (Table 1, entries 7 and 8). Both **3aa** and **3aa′** were observed when $Ni(acac)_2{\cdot}H_2O$ was used as the catalyst. Reaction temperature studies revealed that either increasing or decreasing the temperature reduced the yield (Table 1, entry 9 or 10, respectively). Subsequently, several different solvents, including alcohol, nonpolar and polar solvents, and ether, were further screened (Table 1, entries 11−15). In the absence of an alcohol as the solvent, the formation of **3aa** was nearly negligible, indicating that the solvent played an important role in the reaction. The oxidative addition of $Ni(cod)_2$ and alcohol might be involved in the critical step of the reaction.[16] Finally, different mixed solvents were examined, and the yield was effectively improved using a toluene/MeOH solvent (Table 1, entry 16).

With the optimal conditions established, we explored the substrate scope of this nickel-catalyzed cascade carbocyclization/alkenylation of bisallenes (Scheme 2). First, the scope of 1,5-bisallenes was investigated, and all of the reactions proceeded smoothly to produce the bicyclo[3.2.0]heptanes in moderate to good yields (49−73%, **3aa**−**3ka**). Different protecting groups on the nitrogen, such as sulfonyl, carboxybenzyl, and benzoyl groups, were all suitably employed

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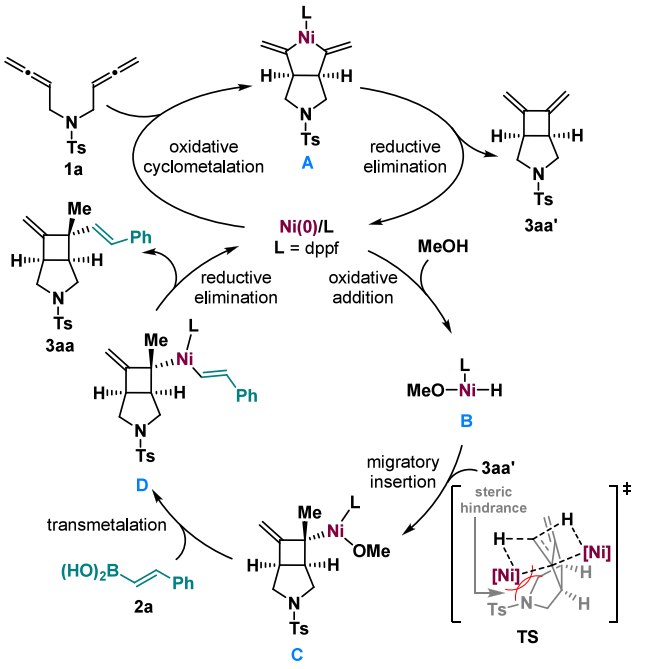
Scheme 2. Substrate Scope of 1,5-Bisallenes and Organoboronic Acids[a]



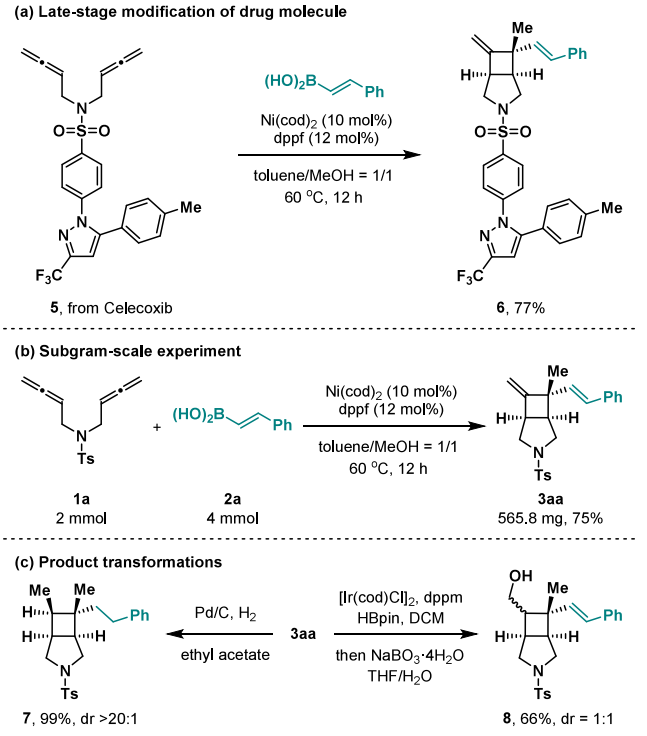
[a]Reactions were conducted with bisallenes **1** (0.2 mmol, 1.0 equiv), boronic acids **2** (0.4 mmol, 2.0 equiv), $Ni(cod)_2$ (10 mol %), dppf (12 mol %), and a toluene/MeOH solvent (1.0 mL/1.0 mL) under a N_2 atmosphere at 60 °C for 12 h. Isolated yields of products **3**. [b]On a 0.1 mmol scale.

in this transformation (**3aa**−**3ia**). The electronic effects of arylsulfonyl groups had no significant impact on the reaction efficiency. Moreover, apart from nitrogen-linked bisallenes, carbon- and oxygen-linked substrates were equally feasible to form the corresponding carbocyclization/alkenylation products in good yields (**3ja** and **3ka**, respectively). When multi-substituted bisallenes **1l**−**1n** were subjected to the standard conditions, they failed to generate either [2 + 2] cycloaddition products or the desired carbocyclization/alkenylation products. Next, the scope of alkeylboronic acids was explored. With a wide range of substituents, including electron-donating and electron-withdrawing groups, at the *para* position of (*E*)-styrylboronic acids, the reactions proceeded to generate the desired products with high regio- and diastereoselectivity (**3ab**−**3ah**). *meta*- and *ortho*-substituted (*E*)-styrylboronic acids were well tolerated, affording the carbocyclization/alkenylation products in moderate to good yields (48−78%,

Scheme 3. Possible Reaction Mechanism



Scheme 4. Synthetic Applications



3ai−**3al**). The aryl chloride might undergo oxidative addition with the nickel(0) catalyst, causing a sharp decrease in yields (**3af** and **3ak**). Besides, thiophene could also be introduced into the bicyclo[3.2.0]heptane skeleton (**3am**). Alkyl-substituted alkenylboronic acids (acyclic and cyclic) were still favorable in this reaction, albeit in relatively low yields (21−67%, **3an**−**3ap**). Furthermore, when kinds of arylboronic acids, including heterocycle-substituted boronic acids, were subjected to the standard conditions, the corresponding carbocyclization/arylation products were obtained in moderate

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to high yields (41−80%, **3aq**−**3av**). However, only [2 + 2] cycloaddition product **3aa′** was formed when alkylboronic acids were used (**2w**−**2y**). Notably, a variety of functional groups, such as halogen, amide, ester, and cyano groups, were found to be compatible under this mild catalytic system. Although the nickel-catalyzed hydroarylation of 1,1- and 1,2-disubstituted alkenes with organoboronic acids has been achieved,[17] even with an excess of boronic acid employed in our reactions, no further hydroarylation or hydroalkenylation occurred at the alkenes of the cyclic product.

Based on our comprehensive mechanistic studies (for details, see Scheme S1) and previous works,[6,17,18] the possible reaction mechanism of this nickel-catalyzed cascade carbocyclization/alkenylation of bisallenes and alkenylboronic acids is shown in Scheme 3. Initially, **1a** undergoes regioselective oxidative cyclometalation with nickel(0)/dppf to yield five-membered nickelacycle **A**. Next, the reductive elimination of complex **A** forms **3aa′**, while regenerating the nickel catalyst. Notably, the nickel-catalyzed [2 + 2] cycloaddition of bisallene **1a** is a fast process, and it would not be obstructed by the addition of **2a**. After the complete conversion of **1a**, the nickel(0) species proceeds to initiate a second catalytic cycle. The oxidative addition of nickel(0) species with MeOH produces nickel hydride complex **B**. Then, coordination and regioselective migratory insertion of **3aa′** with nickel hydride complex **B** result in the formation of allylic nickel complex **C**. The transition state for this process shows that nickel hydride insertion from the front side encounters more steric hindrance than that from the back side. Subsequently, complex **C** undergoes transmetalation with **2a** to yield complex **D**. Finally, complex **D** undergoes reductive elimination to afford **3aa** and regenerate the nickel(0) species. The overall reaction mechanism contains two catalytic cycles. The first is the nickel-catalyzed regioselective [2 + 2] cycloaddition of bisallenes, and the second involves the subsequent hydroalkenylation of the diene intermediate with alkenylboronic acids. This represents a unique cascade process for the transition metal-catalyzed carbocyclization/alkenylation of bisallenes.[12−14]

To demonstrate the synthetic utility of this method, we first performed late-stage modification of a drug molecule (Scheme 4a). When celecoxib-derived bisallene **5** was subjected to the standard conditions, cascade carbocyclization/alkenylation product **6** was successfully obtained in good yield. Significantly, the reaction between **1a** and **2a** could be scaled up to the subgram level, yielding **3aa** with sustained efficiency (Scheme 4b). Furthermore, downstream transformations of **3aa** were conducted (Scheme 4c). Under a hydrogen atmosphere, the 1,1- and 1,2-disubstituted alkenes in **3aa** could be readily reduced to form bicyclo[3.2.0]heptane **7** in quantitative yield with excellent diastereoselectivity. Additionally, in the presence of an iridium catalyst, one-pot hydroboration/oxidation selectively occurred at the 1,1-disubstituted alkene of **3aa**, although with relatively low diastereoselectivity.

In conclusion, we achieved the first nickel-catalyzed cascade carbocyclization/alkenylation of bisallenes and organoboronic acids. Carbon-, nitrogen-, and oxygen-linked bisallenes and alkenyl and arylboronic acids are well tolerated in this reaction, resulting in the formation of various bicyclo[3.2.0]heptane derivatives with high selectivity and efficiency. Mechanistic experiments indicate that this reaction occurs through a unique two-step cascade process. It starts with a nickel-catalyzed [2 + 2] cycloaddition of bisallenes, followed by a nickel-catalyzed regioselective hydroalkenylation of the diene intermediate with alkenylboronic acids. Moreover, the practical utility of this method is showcased through the late-stage modification of drug molecules, scale-up experiments, and downstream transformations. Further investigations of the nickel-catalyzed carbocyclization of bisallenes are ongoing in our laboratory and will be reported in the near future.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c02170.

Experimental procedures, characterizations, and analytical data of new compounds, X-ray diffraction data for **3aa**, and NMR spectra of new compounds (PDF)

Accession Codes

Deposition Number 2452146 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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